

Faisal Yusuf Ughratdar

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EDUCATION

University of Illinois Chicago

Bachelor of Science in Mechanical Engineering, GPA: 3.3/4.0

- Dean's List recipient

Chicago, IL

Expected May 2026

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, AutoCAD, Revit, ANSYS Mechanical/Fluent, SolidWorks Motion, GD&T

Robotics & Controls: ROS 2 Humble, Arduino, Raspberry Pi, CNC Shield, stepper motor control, inverse kinematics

Programming & Data: Python, MATLAB, C++, data analysis, uncertainty analysis, technical documentation

Manufacturing: CNC machining, FDM 3D printing, DFM principles, material selection, prototyping, process optimization

ENGINEERING PROJECTS

Pentagon Robot

August 2025 – May 2026

University of Illinois Chicago, Department of Mechanical Engineering

Chicago, IL

- Designed and fabricated a low-cost five-bar parallel planar pick-and-place robot for ping-pong ball transfer between matrix locations, targeting ≤ 10 second cycle time and ± 0.5 mm placement accuracy within a \$350 prototype budget
- Modeled and validated the full assembly in SolidWorks and ROS 2 Humble/RViz2 using URDF export to verify workspace reachability, singularity avoidance, and link-interference clearance
- Integrated Raspberry Pi 4, Arduino UNO R3/CNC Shield, NEMA 17 stepper motors, vacuum end effector, and pressure-sensor feedback with automatic retry logic for failed pickup attempts

PROFESSIONAL EXPERIENCE

Computer Specialist

January 2023 – May 2026

University of Illinois Chicago, College of Nursing

Chicago, IL

- Deployed and configured 200+ laptops and desktops using Microsoft Intune, enforcing security baselines, software policies, and device compliance for faculty and staff
- Resolved Tier 1 and Tier 2 support requests for 100+ users across networking, VPN, software, printer, and classroom AV issues while maintaining a 95% on-time resolution rate
- Managed imaging, patching, hardware refresh, proactive maintenance, and monitoring workflows, reducing recurring critical failures by 20%

Thermodynamics Laboratory Assistant

Fall 2024

University of Illinois Chicago, Department of Mechanical Engineering

Chicago, IL

- Prepared heat exchanger, combustion, and thermodynamic cycle experiments while guiding students through setup, data acquisition, and uncertainty analysis
- Troubleshoot measurement issues and reviewed lab data for consistency, improving repeatability and technical report quality

LEADERSHIP & ACTIVITIES

Student Ambassador (Volunteer) – UIC Engineering Outreach

Fall 2024

- Led hands-on engineering demonstrations for 500+ prospective students and families during campus recruitment events
- Coordinated event logistics and volunteer teams while explaining engineering projects to promote STEM engagement